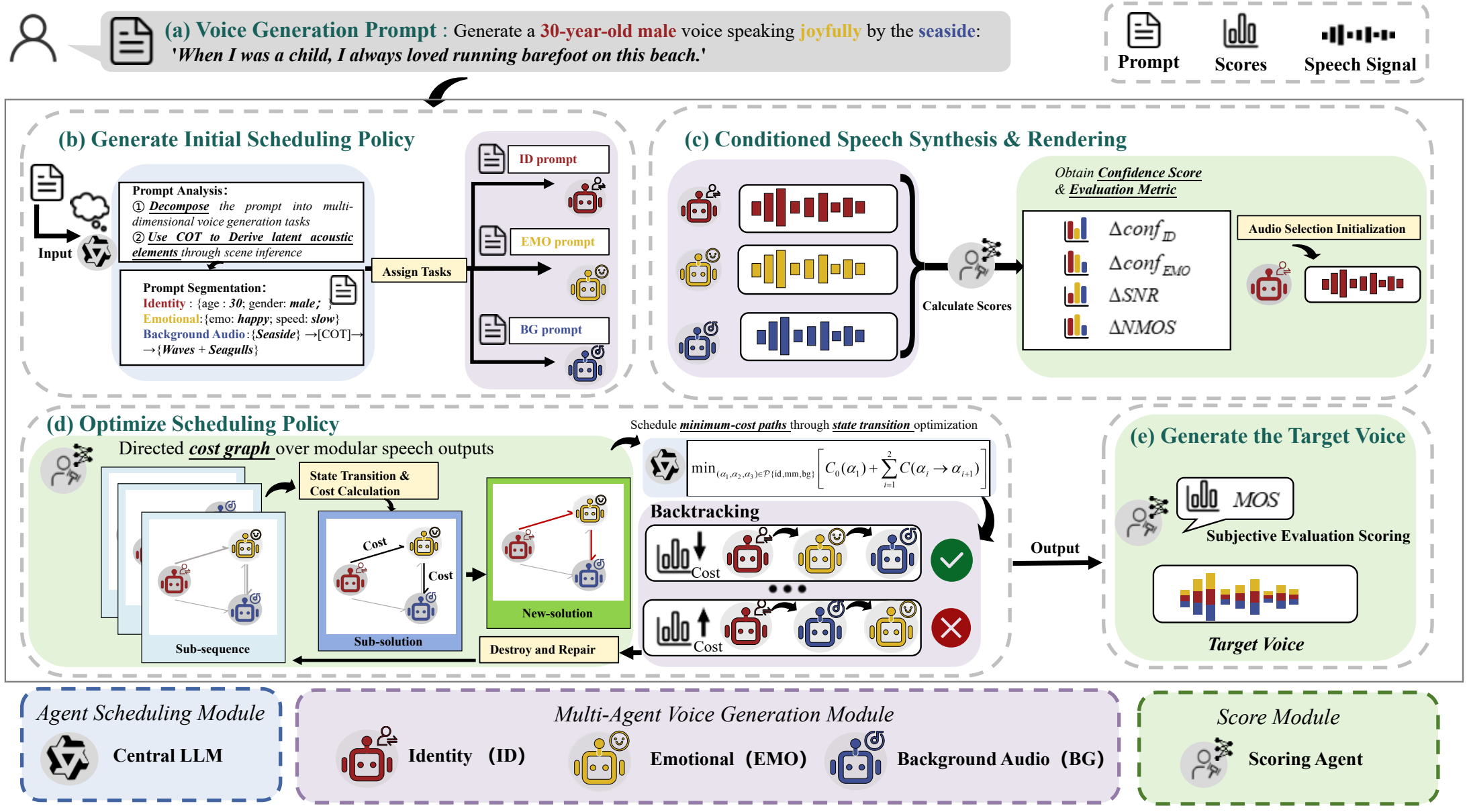
(a) Voice Generation Prompt : Generate a 30-year-old male voice speaking joyfully by the seaside:
'When I was a child, I always loved running barefoot on this beach.'
Prompt
Scores
Speech Signal
(b) Generate Initial Scheduling Policy
Input
Prompt Analysis:
① Decompose the prompt into multi-dimensional voice generation tasks
② Use COT to Derive latent acoustic elements through scene inference
Prompt Segmentation:
Identity : {age : 30; gender: male; }
Emotional:{emo: happy; speed: slow}
Background Audio:{Seaside} →[COT]→
→{Waves + Seagulls}
Assign Tasks
ID prompt
EMO prompt
BG prompt
(c) Conditioned Speech Synthesis & Rendering
Calculate Scores
Obtain Confidence Score & Evaluation Metric
$\Delta conf_{ID}$
$\Delta conf_{EMO}$
ΔSNR
$\Delta NMOS$
Audio Selection Initialization
(d) Optimize Scheduling Policy
Directed cost graph over modular speech outputs
State Transition & Cost Calculation
Sub-sequence
Cost
Cost
Sub-solution
New-solution
Destroy and Repair
Schedule minimum-cost paths through state transition optimization
$\min_{(\alpha_1,\alpha_2,\alpha_3)\in\mathcal{P}\{\mathrm{id,mm,bg}\}}\left[C_0(\alpha_1)+\sum_{i=1}^{2}C(\alpha_i\rightarrow\alpha_{i+1})\right]$
Backtracking
Cost
Cost
Output
(e) Generate the Target Voice
MOS
Subjective Evaluation Scoring
Target Voice
Agent Scheduling Module
Central LLM
Multi-Agent Voice Generation Module
Identity (ID)
Emotional (EMO)
Background Audio (BG)
Score Module
Scoring Agent